

AIGEN IM ENNSTAL, Austria

WMO: 111570

Lat: 47.5328N

Lon: 14.1383E

Elev: 2138

StdP: 13.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	9.2	1.2	6.3	5.6	6.3	8.2	10.8	22.6	41.9	19.4	36.8	1.6	240	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	84.8	66.8	81.4	65.5	78.1	64.1	68.5	81.4	66.6	78.7	65.1	76.0	5.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	96.4	75.5	62.3	90.9	72.6	60.8	86.0	70.3	34.1	81.4	32.4	78.9	31.1	76.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.2	12.2	-1.9	90.7	6.3	2.7	-6.5	92.7	-10.2	94.2	-13.7	95.7	-18.3	97.7		
(5)			-2.3	72.3	6.3	1.8	-6.8	73.6	-10.5	74.7	-14.0	75.7	-18.5	77.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	26.8	31.1	39.0	47.0	55.1	61.7	64.3	63.4	55.5	47.5	37.7	28.6		
	DBStd	14.82	8.51	7.89	6.94	6.53	6.26	6.62	5.54	5.42	5.69	6.33	7.43	7.97		
	HDD50	2927	721	529	343	129	23	3	0	1	16	124	374	664		
	HDD65	6879	1185	949	805	539	311	137	80	94	288	543	819	1129		
	CDD50	1672	0	0	2	40	182	355	445	417	180	46	5	0		
	CDD65	149	0	0	0	0	4	39	59	45	2	0	0	0		
	CDH74	1741	0	0	0	10	120	463	595	505	48	0	0	0		
	CDH80	441	0	0	0	1	21	124	163	128	4	0	0	0		
(14) Wind	WSAvg	4.5	4.5	4.8	5.5	5.0	4.8	4.6	4.5	4.0	4.0	3.9	4.0	4.1		
(15) Precipitation	PrecAvg	53.0	3.4	2.9	4.1	3.1	4.8	6.2	7.0	6.0	5.0	3.7	3.4	3.4		
	PrecMax	62.7	8.4	7.0	8.7	5.5	8.4	10.6	13.3	10.7	9.7	7.2	8.5	8.4		
	PrecMin	42.2	0.4	1.0	1.5	0.9	1.4	3.5	3.0	2.6	2.1	1.5	0.4	1.2		
	PrecStd	4.9	2.0	1.6	2.1	1.1	1.8	1.6	2.6	1.8	1.9	1.5	1.7	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	55.5	64.7	75.0	82.8	88.4	89.3	88.0	78.9	70.1	64.5	49.2		
		MCWB	42.7	44.5	49.6	56.5	62.6	67.5	68.5	67.6	64.2	58.7	51.1	43.6		
	2%	DB	43.5	49.5	60.4	70.0	77.2	83.7	84.4	83.6	74.4	65.5	56.8	44.7		
		MCWB	39.0	41.5	47.9	53.8	60.7	65.5	67.2	67.0	62.6	56.5	49.7	40.2		
	5%	DB	40.1	44.6	55.4	65.8	73.1	79.6	80.7	79.7	70.9	61.7	51.6	41.1		
		MCWB	36.7	39.0	45.4	52.2	58.5	64.1	65.8	65.8	60.4	54.1	46.9	37.8		
	10%	DB	37.3	41.0	50.5	61.3	68.7	75.3	76.7	75.8	67.1	58.3	47.9	38.2		
		MCWB	34.5	36.6	43.3	49.9	56.9	62.5	64.3	64.2	58.5	52.5	44.7	35.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	45.2	50.9	57.8	64.7	69.7	71.2	70.7	65.7	60.1	52.9	44.7		
		MCDB	49.0	53.6	61.7	72.2	78.3	83.7	84.8	83.2	76.5	68.2	61.6	48.2		
	2%	WB	39.5	41.9	48.5	54.9	61.8	67.2	68.4	68.4	63.3	57.3	49.8	40.9		
		MCDB	42.7	48.2	58.4	67.9	74.5	80.5	81.2	80.8	72.9	63.7	56.3	43.9		
	5%	WB	36.8	39.3	46.3	52.7	59.7	65.2	66.5	66.4	61.2	55.1	47.4	38.1		
		MCDB	39.6	43.9	54.4	63.7	70.8	77.3	78.4	77.7	69.0	60.4	51.4	40.5		
	10%	WB	34.8	37.0	43.7	50.5	57.6	63.2	64.8	64.5	59.1	53.0	44.8	35.9		
		MCDB	36.9	40.5	49.9	59.9	67.2	73.5	75.2	74.4	65.9	57.6	47.7	37.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	15.1	17.9	20.7	21.0	20.2	20.2	20.7	19.0	18.5	13.4	11.2		
		MCDBR	13.8	20.8	28.3	31.9	30.6	29.6	28.8	29.0	27.0	24.5	19.8	13.4		
		MCWBR	10.0	13.6	16.7	17.1	15.5	13.4	13.6	14.3	15.6	15.7	13.4	10.0		
		MCDBR	12.8	18.9	26.4	29.1	27.4	26.8	26.3	26.6	24.7	22.6	18.8	12.5		
	5% WB	MCWBR	9.6	12.7	16.0	16.1	14.3	12.8	12.9	13.9	14.7	15.0	12.9	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.313	0.357	0.399	0.410	0.416	0.419	0.411	0.379	0.347	0.316	0.284		
	taud		2.376	2.305	2.260	2.235	2.254	2.282	2.277	2.317	2.381	2.443	2.467	2.441		
	Ebn at Noon		255	268	272	270	271	270	267	264	262	254	240	241		
	Edh at Noon		22	30	36	41	42	41	41	37	32	26	21	19		
(44) All-Sky Solar Radiation	RadAvg		387	671	991	1304	1501	1646	1651	1475	1071	742	415	318		
	RadStd		44	82	112	146	174	145	156	173	139	73	62	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	N/A	N/A	+1.89	+1.01	+0.74	N/A	N/A	N/A	N/A
(47)	Regional (103 neighbors)	+0.92	N/A	N/A	+1.42	+0.83	+0.70	N/A	-250	+146	+77

Nomenclature: See separate page